# Drug Repurposing in Rheumatoid Arthritis

*Section 4 — working draft (Days 1–3)*

## 4.1. Antidiabetic Drugs

Several drugs for type 2 diabetes also act on pathways that drive inflammation in RA, such as AMPK and NF-κB (Section 3). Insulin resistance is also common in RA and is linked to higher disease activity [Ormseth2013]. This section covers metformin and pioglitazone first, then the newer GLP-1 receptor agonists, and finally SGLT2 inhibitors.

### 4.1.1. Metformin

Metformin is an inexpensive first-line drug for type 2 diabetes and is generally safe [Gharib2021]. Its main mechanism is activation of AMP-activated protein kinase (AMPK), which regulates glucose and lipid metabolism, and the drug also has anti-inflammatory effects [Son2014]. AMPK activation inhibits mammalian target of rapamycin (mTOR), which alters T-cell responses; in particular, metformin inhibits the differentiation of Th17 cells, important drivers of chronic inflammation [Son2014].

Animal studies consistently show that metformin reduces arthritis and protects the joint. In the collagen-induced arthritis (CIA) model (Table 1), immunized mice were given 5 mg/mouse of metformin

orally three times per week for 9 weeks, starting 7 days after the first immunization. This treatment significantly reduced arthritis severity and restored the Th17/Treg balance, with fewer Th17 cells and more Treg cells. Phosphorylation of signal transducers and activators of transcription 3 and 5 (STAT3 and STAT5) fell and rose, respectively, and osteoclast activity was reduced [Son2014]. Similarly, in rats with CIA, intraperitoneal (i.p.) metformin (100 mg/kg every 3 days for 3 weeks) protected the knee joints on micro-computed tomography (micro-CT) and on histopathology. Micro-CT showed preserved trabecular bone, histology showed less osteoclast formation and synovitis, and metformin also lowered markers of systemic inflammation, chondrocyte apoptosis and cartilage matrix degradation [Fan2020].

Data from human joint tissue also suggest a link between AMPK and RA activity. In synovium from patients with RA, the active (phosphorylated) form of AMPK-α1 was higher than in synovium from osteoarthritis patients; in their serum, phosphorylated AMPK-α1 rose with the 28-joint Disease Activity Score (DAS28), and total AMPK-α1 rose with tumor necrosis factor-α (TNF-α) [Chen2020]. When fibroblast-like synoviocytes (FLS) from these patients were cultured, metformin at high (millimolar) doses inhibited cell proliferation and migration and reduced TNF-α, interleukin (IL)-1β and IL-6, although the authors questioned whether such doses can be reached in

patients [Chen2020].

Two small clinical studies suggest a benefit of adding metformin to conventional synthetic (cs) disease-modifying anti-rheumatic drugs (DMARDs). A randomized, single-blind trial enrolled 66 Egyptian patients with moderately or highly active RA, of whom 33 received metformin 850 mg twice daily in addition to csDMARDs and 33 received placebo. After 6 months, C-reactive protein (CRP) levels and DAS28-CRP scores were lower than with placebo, and only three metformin-treated patients withdrew because of gastrointestinal side effects [Gharib2021]. In a non-randomized prospective cohort study, 124 patients received low-dose metformin (250 mg twice daily) in addition to traditional csDMARD therapy for 90 days and were compared with 98 patients receiving csDMARDs only. Metformin use was associated with a faster fall in IL-6, TNF-α, high mobility group box 1 (HMGB1) and CRP than in the csDMARD-only group [Zhang2023].

Data on RA prevention are conflicting. In a large UK cohort of people recently diagnosed with type 2 diabetes, RA risk was the same in metformin users as in users of other oral diabetes drugs (hazard ratio [HR] after adjustment 1.00, 95% CI 0.82 to 1.22) [Zemedikun2021]. A Mendelian randomization study suggested that metformin may protect against RA, but its genetic markers reflect metformin use and therefore largely track diabetes [Liang2023]. Therefore,

metformin may be a safe and inexpensive add-on to csDMARDs to lower RA disease activity. Whether it prevents RA is still unclear, and larger and longer double-blind randomized controlled trials (RCTs) that also check joint damage on X-ray are needed.

### 4.1.2. Pioglitazone

Pioglitazone, a thiazolidinedione, activates the nuclear receptor peroxisome proliferator-activated receptor (PPAR)-γ, and although it is mainly used as an antidiabetic drug, it may also have anti-inflammatory effects. PPAR-γ activation suppresses pro-inflammatory genes and the production of inflammatory cytokines, matrix metalloproteinases (MMPs), and reactive oxygen species (ROS) [Shahin2011]. Another reason to test it in RA is that insulin resistance and inflammation are mutually reinforcing [Ormseth2013].

In animals, pioglitazone was effective only at high doses or when combined with methotrexate. In the adjuvant-induced arthritis (AIA) rat model, oral pioglitazone reduced disease severity, but only at the top dose of 30 mg/kg/day [Koufany2008]. It reduced bone erosions and inflammation-associated bone loss but failed to prevent cartilage lesions, and these anti-arthritic doses were higher than those needed to restore insulin sensitivity and increased body fat [Koufany2008]. In a later study using the same model, pioglitazone alone (1.35 mg/kg/day) had no significant effect on disease activity [Roy2017]. Combined with

methotrexate (MTX; 0.225 mg/kg/day), it lowered disease activity more than either drug alone ($P < 0.05$) [Roy2017].

Clinical trials show a small benefit at best. An Egyptian nonrandomized, single-blind trial enrolled 49 patients with RA and type 2 diabetes, all maintained on MTX 15 mg/week [Shahin2011]. The 28 patients with more active disease added oral pioglitazone 30 mg/day for 12 weeks, while the other 21 took placebo [Shahin2011]. At 12 weeks, DAS28 ($P = 0.001$) and CRP ($P < 0.0001$) were lower in the pioglitazone group than in the placebo group, but ESR stayed higher, and weight gain occurred in 25% of treated patients [Shahin2011]. The placebo group also improved from baseline, and the two groups did not start with the same disease activity, which weakens this result [Shahin2011].

In a US crossover trial that was randomized, double-blind and placebo-controlled, 34 patients with RA on stable therapy received pioglitazone 45 mg/day and placebo for 8 weeks each [Ormseth2013]. Pioglitazone produced a modest 9.3% fall in DAS28-CRP ($P = 0.046$) and no significant change in DAS28 using erythrocyte sedimentation rate (ESR; $P = 0.92$), and lower-extremity edema occurred in 16% of participants while on pioglitazone but in none while on placebo [Ormseth2013].

A larger crossover trial with the same design and dose enrolled 143 non-diabetic patients with RA for 3 months

per treatment arm, and DAS28-CRP decreased by a mean of 0.31 on pioglitazone compared with 0.15 on placebo (P = 0.02) [Marder2013]. However, the authors concluded that these findings did not, at that stage, support adding PPAR-γ agonists to the standard treatment of RA [Marder2013]. The benefit in these trials was small and came mainly from CRP-based scores, not from ESR or joint counts [Ormseth2013][Marder2013]. Given the risk of fluid retention and weight gain, pioglitazone should not be used routinely in RA, although it may still be studied as an add-on for patients with RA who also have insulin resistance or type 2 diabetes.

### 4.1.3. GLP-1 Receptor Agonists

Liraglutide, semaglutide and dulaglutide are receptor agonists of glucagon-like peptide-1 (GLP-1RAs), which raise insulin release when glucose is high, lower glucagon, slow gastric emptying and increase satiety [Israel2026]. Two lines of evidence link these drugs to RA. In a cross-sectional study of 151 patients with RA but without diabetes and 210 sex-matched controls, the patients had an impaired incretin-insulin axis, with higher GLP-1, glucose-dependent insulinotropic polypeptide (GIP) and insulin levels and more insulin resistance [TejeraSegura2017]. GLP-1 receptors are also found on immune cells (T and B cells, monocytes and macrophages) [Israel2026]. There, the drugs have been reported to lower TNF-α, IL-6 and IL-1β, block nuclear

factor kappa B (NF-κB) and favor M2 (anti-inflammatory) macrophages [Israel2026].

Clinical data in RA come mainly from a single-center chart review of overweight or obese patients (BMI ≥ 27 kg/m²) [Kellner2025]. It compared 173 patients who took semaglutide (most patients) or tirzepatide (a dual GIP/GLP-1 receptor agonist) with 42 patients who were prescribed one of these drugs but did not take it [Kellner2025]. Disease activity, graded from clinic notes on a 0–3 scale rather than DAS28, changed by −0.03 versus +0.20 in controls ($P = 0.03$), a small difference of unclear clinical meaning [Kellner2025]. Pain fell by 0.6 cm on a visual analogue scale in the treated group but rose by 1.3 cm in controls ($P < 0.001$) [Kellner2025]. ESR and CRP did not fall more than in controls, and the pain relief was not linked to weight loss, but the study was small and the reasons for starting the drug were not measured [Kellner2025].

Not all data indicate a benefit. A case-control study compared 4,535 RA patients with 22,675 controls and found that use of semaglutide or liraglutide in the 10 years before diagnosis was linked to higher odds of new-onset RA (adjusted odds ratios 1.64 and 1.58) [Israel2026]. Dulaglutide showed a similar but non-significant trend (adjusted odds ratio 1.31) [Israel2026]. This link was seen mainly with short use (≤6 months), and the authors suggest the drugs may cause a short-lived rise in inflammation early in treatment [Israel2026]. However, confounding may

explain part of this, because obesity and diabetes, the main reasons for giving these drugs, raise RA risk on their own, and the authors add that longer use may lower inflammation through weight loss and better insulin sensitivity [Israel2026]. The two studies also address different questions, new-onset RA versus established RA, and their results are therefore not necessarily in conflict.

To date, the evidence is limited, observational and mixed, and we found no randomized trials in RA. GLP-1RAs may help some patients with RA and obesity, but they cannot be recommended yet, and future trials should check whether the first months of treatment briefly worsen inflammation.

### 4.1.4. SGLT2 Inhibitors

Sodium–glucose cotransporter 2 (SGLT2) inhibitors, such as dapagliflozin, are oral drugs for type 2 diabetes that block glucose reabsorption in the kidney [ElDemerdash2025]. The only animal study in arthritis that we found reported clear anti-inflammatory effects. In an Egyptian study (Ain Shams University) of AIA rats induced with complete Freund's adjuvant, three oral daily doses of dapagliflozin (1, 5 and 10 mg/kg) were tested for 3 weeks; the highest dose worked best [ElDemerdash2025]. Compared with untreated arthritic rats, dapagliflozin alone lowered serum NF-κB p65 by 66% and TNF-α by 71% [ElDemerdash2025]. It raised the p-AMPK/t-AMPK ratio in the joint by 374%,

whereas MTX alone did not change it, which suggests involvement of the AMPK pathway described for metformin (Section 4.1.1) [ElDemerdash2025]. It also raised joint gene expression of the autophagy markers autophagy-related gene 7 (ATG-7), Beclin-1 and unc-51-like kinase 1 (ULK-1) [ElDemerdash2025]. On X-ray, neither drug alone reduced bone erosion or joint-space narrowing; only the two drugs together did [ElDemerdash2025].

Human data are limited to RA onset in people with diabetes: a study used TriNetX records to emulate target trials in adults with newly diagnosed type 2 diabetes [Yen2026]. It compared SGLT2 inhibitors with sulfonylureas, dipeptidyl peptidase-4 (DPP-4) inhibitors or pioglitazone (more than 80,000 matched pairs per comparison) [Yen2026]. In the first year, SGLT2 inhibitor use was linked to a slightly lower risk of new RA than sulfonylurea use (0.118% vs 0.131%; HR 0.90, 95% CI 0.84–0.97), whereas the differences against DPP-4 inhibitors (HR 0.91) and pioglitazone (HR 0.92) were not significant [Yen2026]. The gap with sulfonylureas shrank with longer follow-up and lost significance, as it also did in a per-protocol analysis [Yen2026]. The authors suggest that one year may be too short, because RA has a long silent phase before diagnosis [Yen2026].

Unlike GLP-1RAs (Section 4.1.3), SGLT2 inhibitors were not linked to a higher RA risk than other glucose-lowering drugs [Yen2026]. Still, we found only

one animal study and one observational study and no trial in patients who already have RA [ElDemerdash2025][Yen2026]; their use in RA therefore remains experimental.

### 4.2. Lipid-Lowering Drugs

Because patients with RA have a higher cardiovascular risk, lipid-lowering drugs are a logical choice to test [Kitas2019]. Statins and fenofibrate, the two classes covered in this section, also act on immune cells and may therefore benefit the joints as well as the heart.

#### 4.2.1. Statins

Statins are widely used in cardiovascular therapy and act mainly as competitive inhibitors of 3-hydroxy-3-methylglutaryl-CoA reductase (HMG-CoA reductase), which reduces de novo cholesterol synthesis [Ren2023]. Statins also have effects other than lowering lipids: by reducing isoprenoid synthesis, they limit the isoprenylation of small guanosine triphosphate (GTP)-binding proteins such as Ras and Rho [Chodick2010][ElBarbary2010]. They also suppress interferon-γ-induced major histocompatibility complex (MHC) class II expression, which reduces T-cell activation [Chodick2010], a key driver of joint inflammation in RA [Xing2015].

Animal studies of statins are mostly positive, but not all. In a preclinical study, mice with serum-induced arthritis were given atorvastatin, pravastatin or simvastatin

intravenously (0.2 mg/kg on days 3, 5 and 7). Atorvastatin and pravastatin reduced paw swelling and knee destruction on histopathology, but simvastatin had no significant effect [Walker2017].

As Vandebriel et al. point out, most earlier CIA studies, which often used high intraperitoneal doses, found statins protective [Vandebriel2012]. However, when they gave CIA mice oral atorvastatin at 0.2 mg/day or oral pravastatin at 0.8 mg/day, arthritis started earlier, and every treated mouse became arthritic, against 7 of 12 controls [Vandebriel2012]. To our knowledge, no other group has reported harm [Vandebriel2012].

Results in patients are mixed. Two meta-analyses showed that statins reduced disease activity measured by DAS28, with standardized mean differences (SMD) of −0.55 (95% CI −0.83 to −0.26) and −0.78 (95% CI −1.08 to −0.49) [Xing2015][Ren2023]. However, the included trials were small and mostly conducted at single centers [Xing2015], and both meta-analyses reported high heterogeneity [Xing2015][Ren2023]. In an unblinded randomized study conducted in Egypt, 30 patients with early RA were divided into two groups of 15 and treated for 6 months with MTX and prednisone, either alone (group 1) or with atorvastatin (group 2). DAS28 was significantly reduced from 6.09 to 3.9 in group 2 compared with 6.19 to 5.27 in group 1 [ElBarbary2010]. In contrast, the largest trial, the double-blind Trial of Atorvastatin for primary prevention of Cardiovascular Events in RA (TRACE

RA), enrolled 3,002 UK patients for cardiovascular end points and found no difference in DAS28 between patients who received atorvastatin and those who received placebo [Kitas2019]. Atorvastatin did lower CRP, but the authors judged this fall too small to matter clinically in RA [Kitas2019]. In short, statins appeared to lower RA disease activity in small trials, but the largest trial did not confirm this. For now, statins in RA are used to lower lipids and cardiovascular risk, as in other patients, and trials with disease activity as the main end point are needed before they can be used to treat RA itself.

### 4.2.2. Fenofibrate

Fenofibrate activates PPAR-α and is mainly used to treat high triglycerides [Zhou2012]. In mouse T cells, fenofibrate (5–20 μM) inhibited Th17-cell differentiation in a dose-dependent manner, by 74% at 20 μM, whereas other PPAR-α agonists (WY14643, GW7647, and bezafibrate) had no such effect, which suggests that fenofibrate acts here through another pathway [Zhou2012]. In AIA rats, fenofibrate (300 mg/kg/day for 12 days) lowered the arthritis score from 11.0 to 5.83, a fall of about 47% ($P < 0.01$) [Castillero2012]. Clinical data on fenofibrate in RA are very limited, and to our knowledge no placebo-controlled trial has yet tested whether the effects seen in animals also benefit patients with RA.

## 4.3. Angiotensin II Receptor Blockers

The renin–angiotensin system (RAS) regulates blood pressure and fluid balance, and its overactivity is a common cause of hypertension. Several studies suggest that the RAS also contributes to RA, particularly through angiotensin-converting enzyme (ACE), which is overexpressed in the peripheral blood and synovium of patients with RA [Wang2013].

Angiotensin II acts mainly on angiotensin II type 1 receptors (AT1R). Through these receptors, it promotes ROS production and activates NF-κB, which then increases the release of several pro-inflammatory cytokines [Mostafa2020]. Angiotensin II type 2 receptors (AT2R) oppose these actions, and their stimulation has been linked to anti-inflammatory and tissue-protective effects [Wang2013].

Animal and cell studies of losartan are largely consistent. In AIA rats, losartan at 10 and 15 mg/kg significantly lowered the polyarthritis index compared with vehicle, although this effect was smaller than that of MTX [Wang2013]. Losartan also shifted the receptor balance in the spleen and synovium towards AT2R, with less AT1R and more AT2R, a change that paralleled the fall in the polyarthritis index [Wang2013]. In a second study in AIA rats, losartan lowered serum IL-6 and TNF-α compared with the untreated arthritic group; these changes were accompanied by a rise in serum glutathione (GSH) and a fall in serum malondialdehyde (MDA), two markers of oxidative stress [AbdelElGaphar2018]. An in vitro study tested

losartan, valsartan, and enalapril on peripheral blood mononuclear cells (PBMCs) collected from 16 patients with RA [Cardoso2018]. Losartan lowered four cytokines, IL-6, IL-17F, IL-22, and interferon-γ (IFN-γ), whereas valsartan and enalapril had few or no effects, which suggests that this action may not be shared by valsartan [Cardoso2018]. However, only one high concentration (100 μM) was tested, and cells from patients with active disease responded less [Cardoso2018].

The main clinical evidence for this class in RA comes from a trial of another angiotensin II receptor blocker (ARB), candesartan. In this Egyptian single-blind, randomized, placebo-controlled trial, 45 patients with RA (15 per group) were treated for 3 months with add-on candesartan (8 mg/day), atorvastatin (20 mg/day), or placebo [Mostafa2020]. Compared with the control group, candesartan significantly lowered serum IL-1β, TNF-α, and CRP and improved DAS28 and the disability index of the Health Assessment Questionnaire (HAQ-DI), whereas ESR was reduced only relative to baseline [Mostafa2020]. Atorvastatin lowered both MDA and rheumatoid factor (RF), but candesartan changed neither significantly [Mostafa2020]. These findings should be interpreted with caution, because patients were grouped according to their other medical conditions, only 45 of 59 eligible patients were analysed, the groups differed at baseline in several variables, including CRP and HAQ-DI, and follow-up

was short [Mostafa2020].

ARBs, particularly losartan, may therefore be a reasonable choice for patients with RA who also need treatment for hypertension, although larger and longer trials of losartan are required to establish any disease-modifying effect.

### 4.4. Emerging Candidates

Several other drugs have been proposed for RA, but their evidence is weak, indirect or mixed (Table 2). In rats with CIA, the non-selective β-blocker propranolol reduced inflammation in the heart and made arrhythmias less likely [Lin2020]. However, its effect on the joints was small: paw swelling and clinical scores stayed close to those of untreated CIA rats and improved only mildly late in the disease [Lin2020]. In pooled data from five trials of tocilizumab and/or csDMARDs (5,502 RA patients), the 594 (10.8%) who used β-blockers, mostly β1-selective ones, reached remission on the Simplified Disease Activity Index (SDAI) less often (adjusted HR 0.70, 95% CI 0.57–0.87) [Abuhelwa2021].

Data for the psychiatric drugs are weaker still. Sertraline, a selective serotonin reuptake inhibitor (SSRI), had no effect on synovitis, cartilage damage or bone erosion in CIA rats, unlike MTX, infliximab and tocilizumab in the same study [Chmielewski2026]. Valproate has been proposed because it inhibits histone deacetylases (HDACs), and two selective inhibitors of

one isoform (HDAC6) lowered arthritis scores in CIA mice [Oh2017]. However, these inhibitors differ from valproate, and we found no clinical study of valproate in RA.

For DPP-4 inhibitors, the data concern RA risk rather than treatment. A nested case-control study of diabetic patients starting second-line therapy (790 RA cases and 11,850 controls) found that, compared with other second-line antidiabetic drugs, DPP-4 inhibitor use was not linked to new RA after adjustment (adjusted OR 1.16, 95% CI 0.94–1.43) [Kathe2018]. In patients who did not use insulin, the odds were higher (OR 1.30, 95% CI 1.001–1.985) [Kathe2018]. We found no RA-specific study of allopurinol, and for now these drugs remain hypotheses or have mixed results; better animal studies should come before any trial in patients with RA.